ANALYTICAL CHEMISTRY CUMULATIVE EXAM

September 18, 1993

This cume is to test your knowledge of the standard addition method. In this example a ion selective electrode (ISE) for chloride ion is used to determine the $[Cl^-]$ in an unknown by standard addition. You may assume that the electrode has a Nernstian response to the analyte and the potential of the electrode is given by:

$$E = K + 0.0591 \log[Cl^-]$$

Where: K is a constant, and $[Cl^-]$ is the chloride concentration of the solution that the ISE is emersed in.

Standard addition procedures are usually performed by laboratory technicians with little theoretical training. The procedure they are given is the following: 10mL of standard added to 100 mL of sample. The potential is measured for the sample and then with the standard added to the sample. The technician then calculated the ΔE cause by the standard. He then goes to a table where he looks up the "Q" value to match the ΔE measured. Q multiplied times the concentration of the standard gives the concentration in the sample.

As an example, suppose the electrode is placed in 100 mL of sample and a reading of 228 mV is obtained. Then 10 mL of standard containing 100 ppm of Cl^- is added and the new reading of 210 mV is observed. Since ΔE = 18 mV, Q = 0.082 from the table. Therefore, the original concentration of Cl^- was (0.082) (100 ppm) = 8.22 ppm.

(50 points) 1. Derive the theoretical equation for $C_x = QC_s$:

$$C_x = \left(\frac{1}{(10 + 1)(10^{\frac{\Delta E}{0.0591}}) - 10} \right) C_s$$

Where C_x is the chloride concentration in the sample
C_s is the chloride concentration in the standard
Q is the value in the table.

(10 points) 2. Calculate 5 values of Q given 5 values of ΔE chosen randomly from 1 mV to 30 mV.

(20 points) 3. If 1 mV is the smallest change measurable and 30 mV is the largest, what is the dynamic range for $[Cl^-]$?

(20 points) 4. What would the equation for Q have to be if a Ca^{2+} ISE was used for a calcium ion analysis?